

B Tech: Second Semester

Session: Feb-May 2025

Subject Name: Engineering Mathematics II (4-0-0-4-4)

Subject Code: UE24MA141B

MATLAB - PRACTICE PROBLEMS:

Unit 1:

Random Variables and Probability Distributions:

1. A sequence of 10 Bernoulli trials with $p = 0.2$, what is the probability of getting exactly 2 successes

output: probability = 0.3020

2. Find the Bernoulli distribution and the binomial distribution for $n = 1$ and $p = 0.5$

output:

bernoulli_pmf =

0.5000 0.5000

binomial_pmf =

0.5000 0.5000

3. If the lifetime of a machine follows an exponential distribution with a mean lifetime of 5 years, calculate the probability that the machine lasts more than 3 years.

output: probability = 0.5488

4. For a probability density function $f(X) = \lambda e^{-\lambda x}$ ($x \geq 0$), show that the total probability is 1

output: total_probability = 1

5. Compare Probability Density Function (PDF) of a Poisson and Normal distribution with parameter $\lambda=5$ and plotting of PDF with bars of width 1 using MATLAB

output:

